

Electrically controlled nano-OLED metasurfaces

Jung-El Ryu & Jeehwan Kim



A nanostencil lithography technique enables fabricating arrays of green-emitting OLEDs with pixels as small as 100 nm and an external quantum efficiency of 13.1%.

For decades, organic light-emitting diodes (OLEDs) have contributed significantly to the advancement of modern display and solid-state lighting technologies, combining mechanical flexibility, thin-film integration, and precise colour tunability. However, despite their technological maturity and widespread industrial adoption, the size of OLEDs has remained largely confined to the micrometre scale. The limited miniaturization of pixels beyond a few micrometres has hindered their integration into nanophotonic systems, where sub-wavelength control of light is essential. Moreover, the organic nature of OLED materials poses challenges for integration with nanophotonic or silicon-based photonic platforms, owing to their chemical sensitivity, thermal instability, and incompatibility with conventional high-temperature or vacuum fabrication processes. Achieving such control requires a fabrication strategy that can pattern fragile organic layers without degrading their luminescent efficiency or charge-transport characteristics^{1,2}.

Traditional microfabrication techniques fall short of meeting this requirement. Photolithography, the cornerstone of semiconductor scaling, relies on solvents, photoresists, and plasma chemistries

that can degrade organic emissive materials. Fine metal mask (FMM) shadow deposition, the industrial standard for OLED patterning, mitigates chemical damage. However, FMM has its own limitations, including mask sagging, poor alignment accuracy, and an inherent resolution limit of several micrometres. Even with mechanical reinforcement or laser-perforated masks, FMM processes rarely exceed ~1,000 pixels per inch (ppi) resolution^{1,2}. Consequently, OLED displays remain far above the optical diffraction limit, rendering their integration with nanophotonic or metasurface architectures largely impractical³.

Now, writing in *Nature Photonics*, Marcato, Oh and colleagues demonstrate a resist-free, self-aligned nanostencil lithography for the direct nanopatterning of OLED emissive layers⁴. The team demonstrate green-emitting nano-OLED pixels as small as 100 nm, with densities approaching 10⁵ ppi and an average external quantum efficiency (EQE) of 13.1%.

The key advance lies in the ability to directly pattern OLED emissive layers at the nanometre scale, an accomplishment previously thought incompatible with fragile organic materials^{5,6}. As illustrated in Fig. 1a, the process begins with the fabrication of a 30–50 nm thick SiN_x membrane patterned with an array of nanometric apertures created through electron-beam lithography. This membrane serves as a reusable nanostencil, suspended a few micrometres above the target substrate. During device fabrication, oxygen plasma etching is performed through the stencil, precisely removing material from the exposed regions of the organic stack. Subsequent thermal evaporation

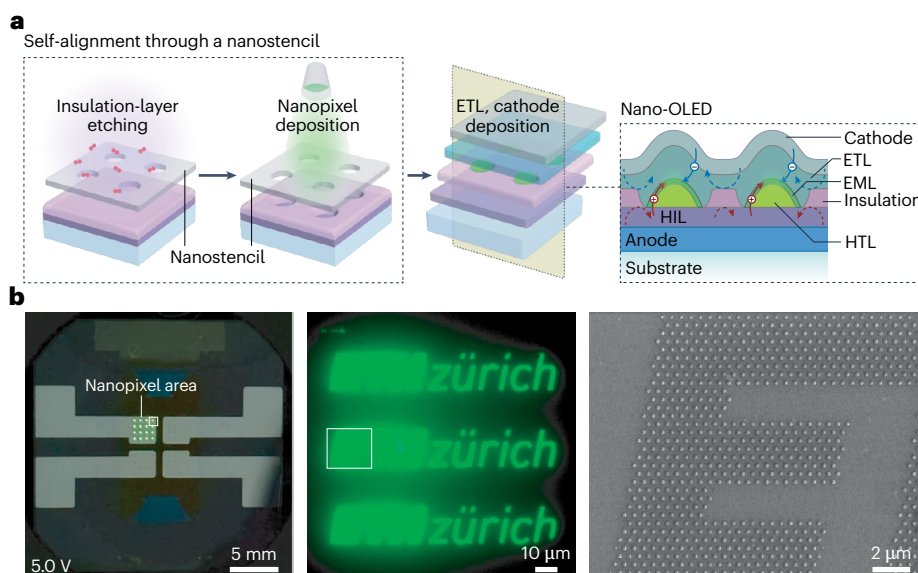


Fig. 1 | Resist-free nanostencil lithography and resulting nano-OLED arrays. **a**, Schematic illustration of the resist-free, self-aligned nanostencil lithography process. Oxygen plasma etching and nanometric-aperture evaporation through

a SiN_x membrane define sub-100-nm OLED pixels without wet processing. **b**, Scanning electron and optical micrographs of the resulting nano-OLED arrays, exhibiting uniform emission and pixel densities approaching 10⁵ ppi.

of emissive and charge-transport layers through the same apertures defines nanoscale pixels without resorting to photoresist coating or wet processing.

Because both etching and material deposition are carried out through the same nanoscale apertures, the process is inherently self-aligned, ensuring precise registration between the etched patterns and deposited pixels. The OLED stack consists of an ITO anode, a PEDOT:PSS hole injection layer, a 40-nm TAPC hole transport layer, a 20-nm Ir(ppy)₃:CBP emissive layer, a 50-nm B3PymPm electron transport layer, a 2-nm Liq electron injection layer, and a 70-nm Al cathode. The resulting nanostructures exhibit uniform thickness, complete isolation between neighbouring pixels, and smooth interfaces undamaged by plasma, as shown in Fig. 1b. Optical and electrical characterization confirm that each 100-nm pixel maintains high emission efficiency, with the overall device sustaining an EQE of 13.1%. Importantly, the pixel array density reaches up to 10⁵ ppi, which is nearly two orders of magnitude higher than previous micro-OLED demonstrations^{5,6}.

The experimentally observed optical behaviour of the nano-OLED arrays is characterized by markedly altered emission directionality and polarization compared with planar OLEDs. Hexagonal nanodisk arrays concentrate light within narrow $\pm 15^\circ$ lobes, bullseye-type concentric arrays yield omnidirectional uniform emission, and one-dimensional nanorod arrays enable reversible modulation of linear polarization (Stokes $S_1/S_0 \approx \pm 0.4$) by tuning the lattice period. These distinct optical responses can be harnessed to engineer high-brightness microdisplays with tailored angular emission for augmented-reality (AR) and virtual-reality optics, to design on-chip optical interconnects or visible-light communication systems requiring directional and polarization-controlled light sources, and to develop polarization-encoded light-emitting metasurfaces for photonic computing and quantum information technologies.

The underlying physical mechanism responsible for these effects originates from collective diffractive coupling among sub-wavelength emitters. When the pixel pitch falls below $\lambda/2$, emission transitions from incoherent, pixel-by-pixel radiation to a regime dominated by interference and near-field coupling. Unlike previous metasurface-driven OLEDs³, where optical modulation relied on patterned electrodes or cavities external to the emissive region, the present approach directly patterns the emissive layer itself, enabling active control of emission through the cooperative coupling of electrically driven emitters. This integration removes alignment complexity, ensures uniform current injection, and achieves sub-wavelength pixel densities without sacrificing efficiency.

Compared to prior approaches, this work represents a substantial conceptual and technical advance. Prior metasurface-based OLEDs relied on nanopatterned electrodes or cavities to control far-field emission but left the emissive layers continuous and unstructured^{3,7}. Likewise, dewetting-assisted templating achieved sub-micron ordering through self-assembly, yet lacked deterministic control and electrical isolation⁸. By contrast, the present nanostencil method directly patterns the emissive organic layers, ensuring sub-100-nm precision, device uniformity, and integration compatibility with standard OLED stacks. The technique promises scalability and solvent-free processing, enabling pattern transfer over millimetre-scale areas and multiple reuses of the same stencil membrane^{5,6}.

Beyond its immediate technological impact, this development positions OLEDs as a bridge between electronics and photonics. In principle, coupling such nano-OLED metasurfaces with CMOS

backplanes could yield ultrahigh-resolution, direction-controlled displays for AR optics¹, or low-power optical transmitters for on-chip communication⁹. The method's compatibility with flexible substrates further suggests applications in conformal photonic skins and wearable nanophotonic emitters.

Despite these promises, several challenges remain before nanostencil-patterned OLED metasurfaces reach full technological maturity. First, the current demonstration employs single-colour (green) emission. Extending the approach to red and blue phosphorescent systems will require careful optimization of deposition rates, thermal budgets, and interlayer adhesion, as these materials are more chemically and thermally fragile². Second, large-area scalability is constrained by the mechanical stability of the nanostencil membrane. Sub-100-nm apertures over centimetre-scale areas are prone to deformation and alignment drift. Advances in membrane materials, such as low-stress SiN_x or graphene-reinforced composites, could mitigate these issues⁶. Third, while the study demonstrates collective optical control, individual pixel addressability has not yet been realized. Integrating active backplanes or hybrid pixel-level electrodes could combine nanoscale emission control with independent electronic modulation.

Looking ahead, the convergence of organic optoelectronics, nanofabrication, and photonic design promises a new generation of light sources that are compact, tunable, and versatile. The concept of electroluminescent metasurfaces may soon expand beyond OLEDs to perovskite, quantum-dot, or hybrid organic–inorganic systems, enabling coherent light generation, optical phase modulation, and even lasing within flexible thin-film devices^{3,9,10}. More broadly, this work exemplifies a growing paradigm in photonics, promoting the integration of emitter, cavity, and metasurface, whereby the control of light becomes as much a matter of structural geometry as of material composition. The nanopatterning technique is thus not only a technical milestone, but a conceptual contribution toward the programmable photonics of organic matter.

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Competing interests

The authors declare no competing interests.